CS560 - UI/UX Design & Implementation

4-1 Activity: Translating Design into Code

Malgorzata Debska

Southern New Hampshire University

Nathan Palmer

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Implementation and Development Process

For this activity, I implemented the Audio Track component in Vue by using the provided Figma prototype as my main guide. I first reviewed the prototype carefully to understand how the final application should look and behave. I looked at the layout, colors, spacing, typography, button placement, and the different interaction states shown in the design. The prototype showed the default screen, the audio playing state, a heart button selected, and multiple heart buttons selected. These examples helped me understand what needed to happen in the Vue application.

The Audio Track component was built to display a list of nature soundtracks. Each track includes the track title, artist name, Heart button, and Play/Pause button. To make the track list dynamic, I used Vue templating and directives. Instead of hard-coding each track separately, I stored the track information in an array and used v-for to display each item. This made the code cleaner and easier to update because new tracks could be added to the array without changing the main layout. According to Garofalo (2025), using consistent and reusable design structures improves both maintainability and user experience because interfaces become easier to understand and manage.

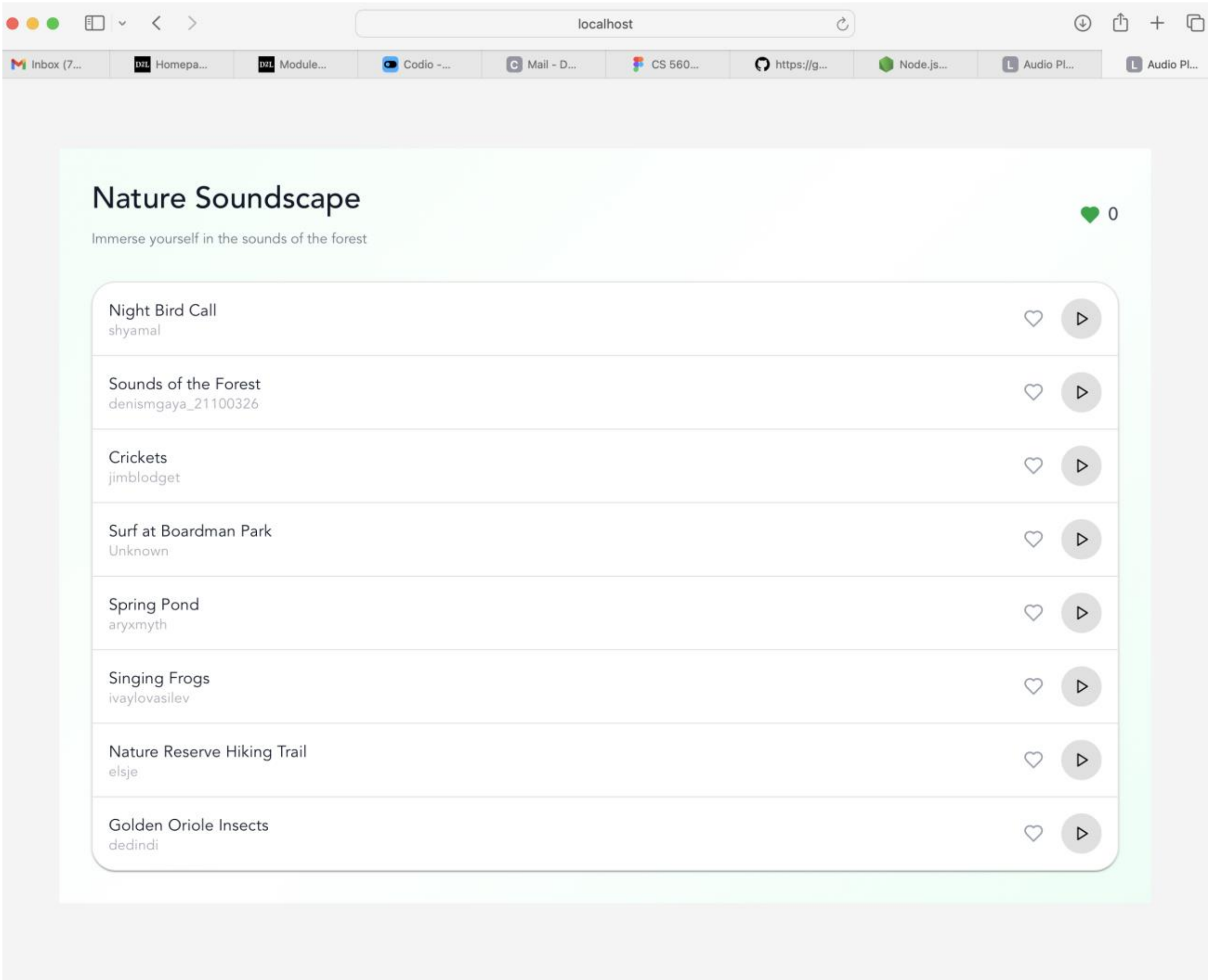
I also used Vue reactivity primitives to manage the application state. The reactive state keeps track of which audio track is currently playing and which tracks have been liked with the Heart button. When a user clicks the Play button, the application updates the active track and changes the button from a gray Play icon to a green Pause icon. If the user clicks on another track, the previous one stops and the new one becomes active. This helped the application match the Figma prototype's audio playing interaction.

The Heart buttons were also connected to a reactive state. When a user clicks a Heart button, the heart changes from an outline style to a filled green style. At the same time, the global heart counter at the top of the screen updates automatically. If the user clicks the same heart again, it becomes unselected and the counter decreases. This shows that the application state is updating correctly based on user interaction.

CSS styling was used to make the Vue application match the Figma prototype. I styled the background with a soft green color, used a white rounded card for the track list, and added spacing between each track row. I also matched the general typography by making the main title larger and darker, while the artist names and subtitle text were smaller and lighter. The Play/Pause and Heart buttons were styled so users could clearly see when an item was active or inactive. This helped maintain visual consistency between the prototype and the final application. Garofalo (2025) explains that typography, spacing, and color are important parts of visual design because they help create hierarchy, improve readability, and provide clear interaction feedback for users.

To verify that the application worked correctly, I tested each required feature in the browser after running the Vue project. I checked that every track title and artist name displayed correctly. I clicked the Play/Pause buttons to make sure the correct icon and color appeared when a track was active. I also tested clicking different Play buttons to confirm that only one track appeared active at a time. Then I tested the Heart buttons by selecting one heart, selecting multiple hearts, and clicking hearts again to remove them. I confirmed that the global counter increased and decreased correctly.

Throughout development, I continuously compared the running Vue application with the original Figma prototype to ensure the design and interactions stayed consistent. Workflows and collaboration between designers and developers are important because they help ensure that the final application accurately reflects the intended user experience and visual design. Overall, my process involved reviewing the Figma design, building the Vue component structure, adding reactive state, using Vue directives to display the dynamic list, applying CSS to match the prototype, and testing all interactions. This helped ensure that the final Vue application matched both the visual design and the expected interactive behavior shown in the Figma prototype.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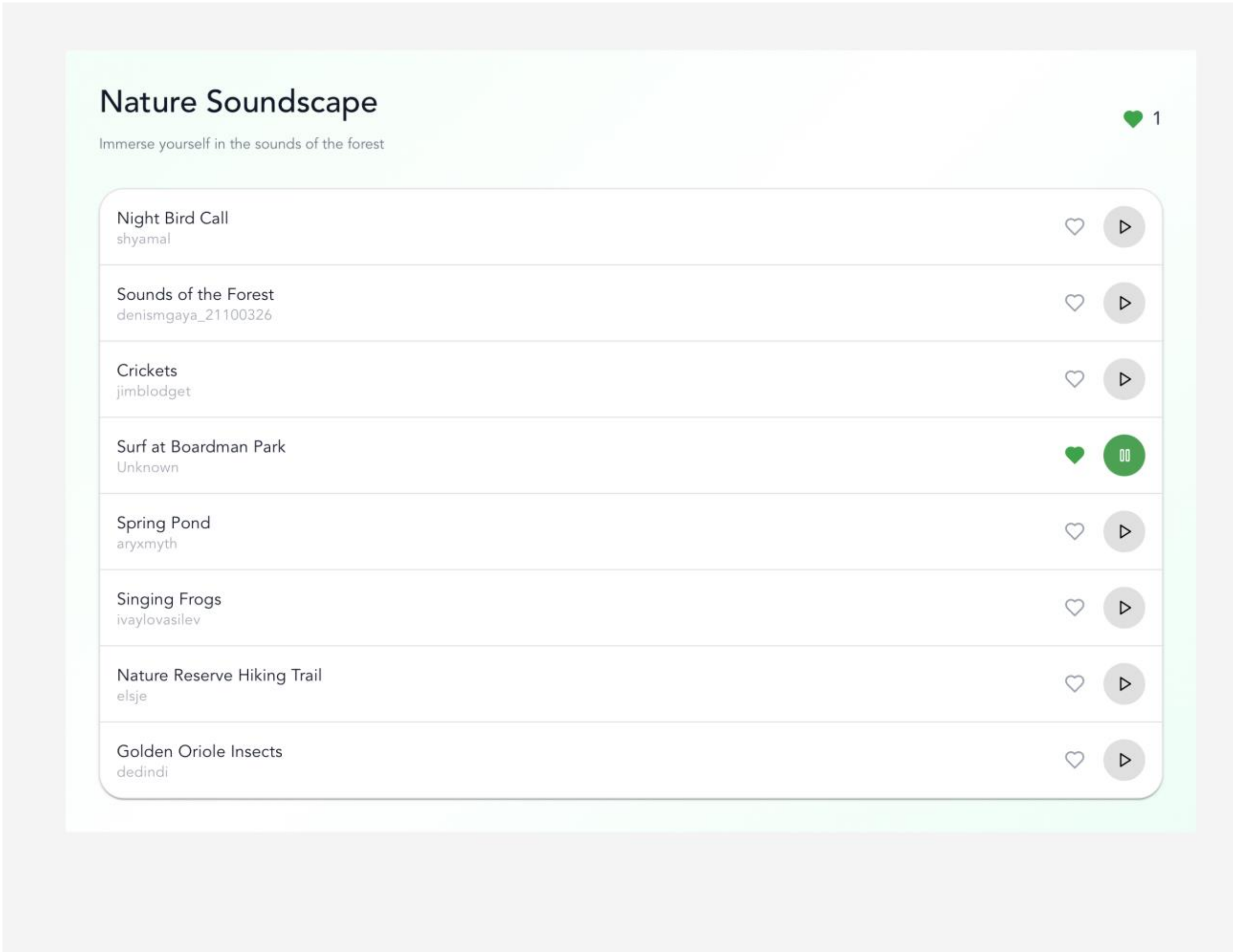
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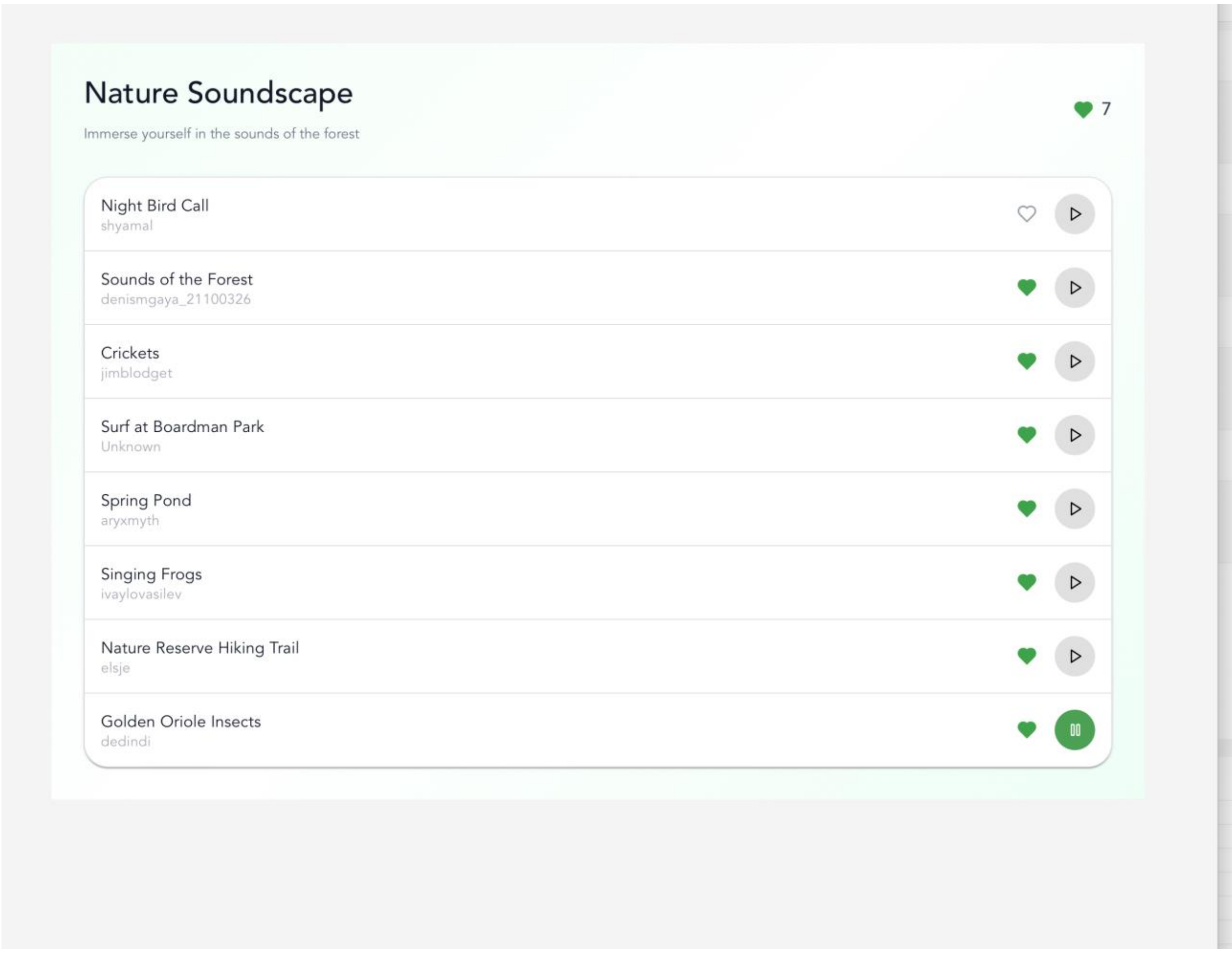
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References

Garofalo, K. (2025). *User Interface/User Experience Design* (1st ed.). Mujo Learning Systems Inc. ISBN-13: 978-1-998671-17-5.